

Architecture Diagrams

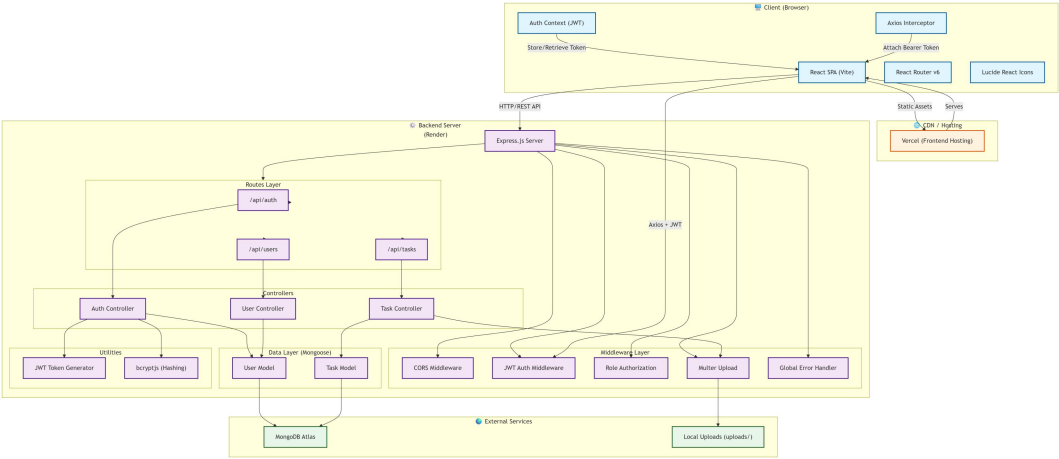
This section provides visual representations of the TaskFlow system architecture, data flow, and key processes. The diagrams illustrate how different components interact and how data moves through the system, from user authentication to task management and file uploads.

Overview of Diagrams

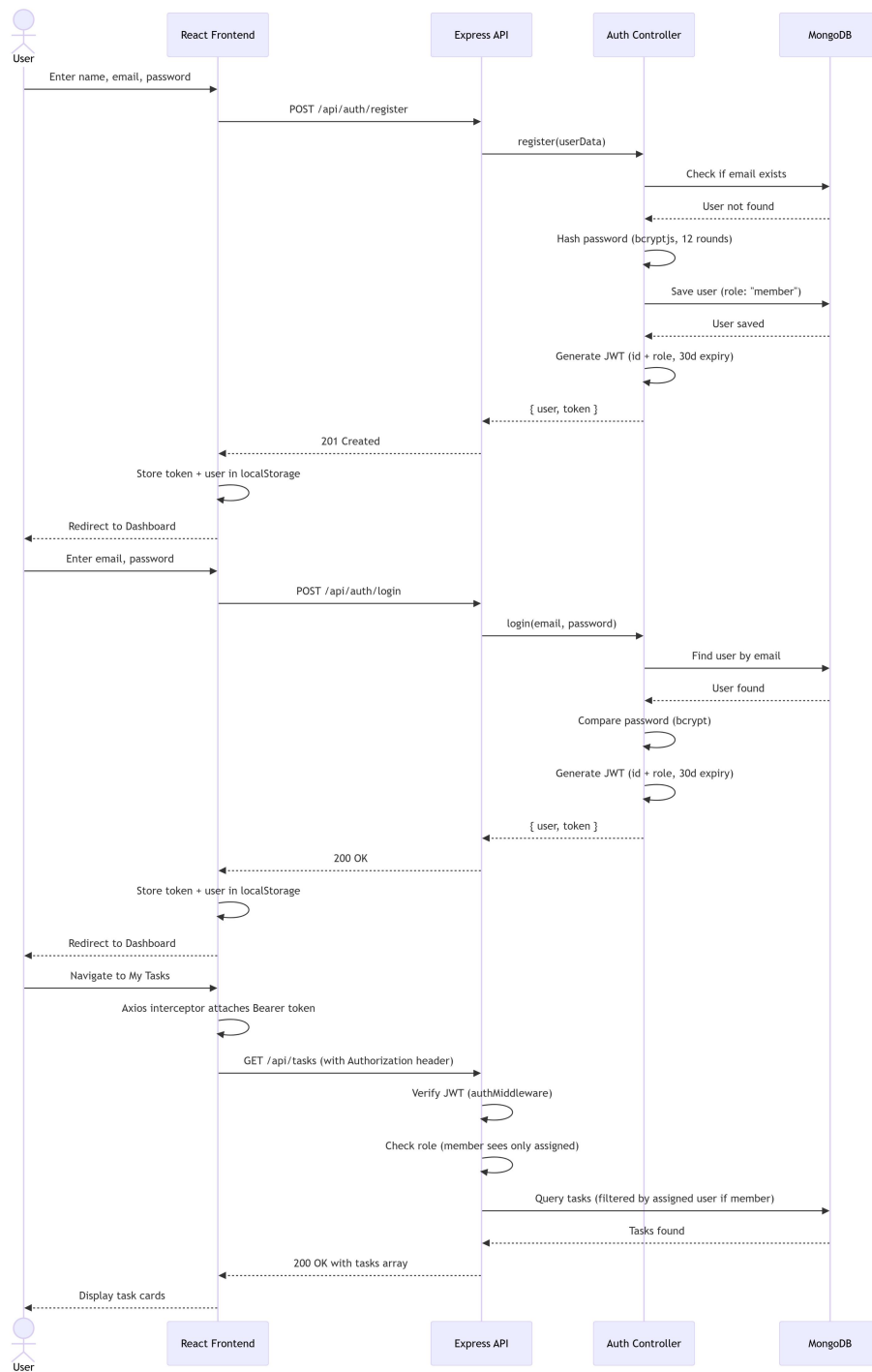
The following diagrams are included in this document:

Diagram	Description
System Architecture	High-level view of all components and their relationships — React frontend, Express backend, MongoDB database, and external services
Authentication Flow	Sequence of steps for user registration, login, and protected route access with JWT tokens
Task CRUD & File Upload Flow	How tasks are created, updated, and deleted, including the complete file attachment process
File Upload Flow (Detailed)	In-depth view of file validation, storage, metadata handling, and download retrieval
File Storage & Retrieval Architecture	Simplified architecture showing the relationship between upload path, disk storage, and MongoDB metadata
Role-Based Access Control Flow	Decision tree showing how Admin and Team Member permissions are enforced
Data Flow	Complete request/response cycle from the React client through Express middleware to MongoDB and back
Deployment Architecture	Production infrastructure on Vercel (frontend), Render (backend), and MongoDB Atlas (database)

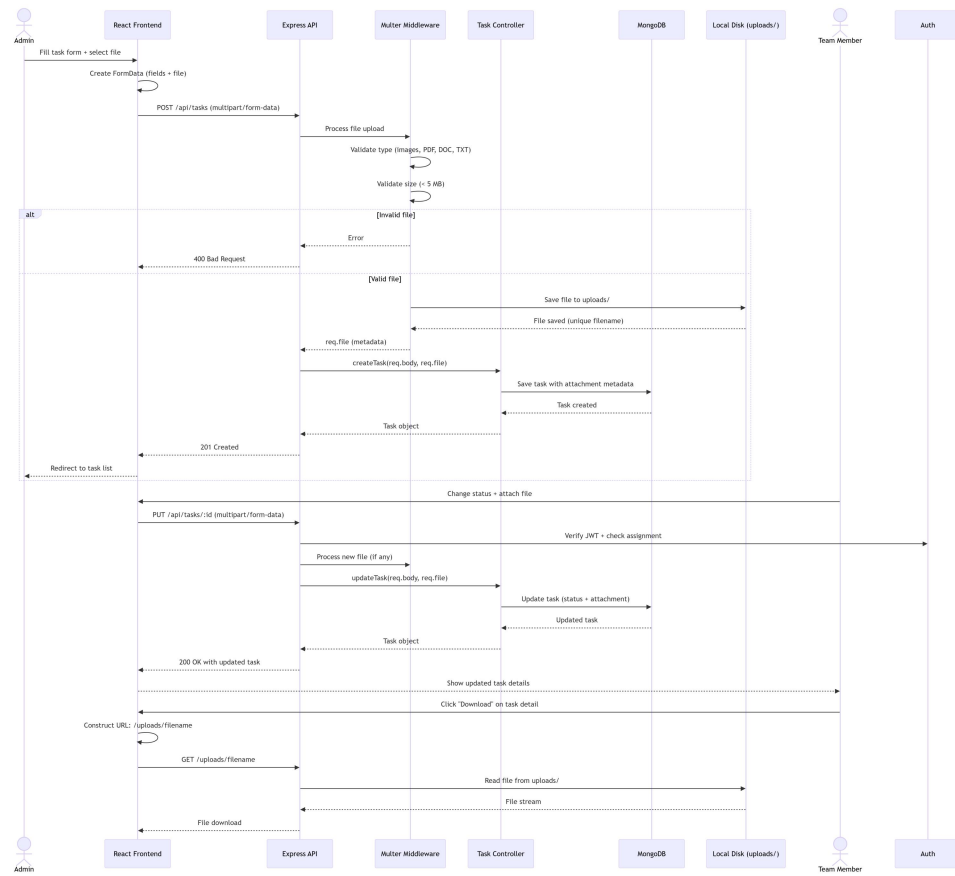
System Architecture Diagram



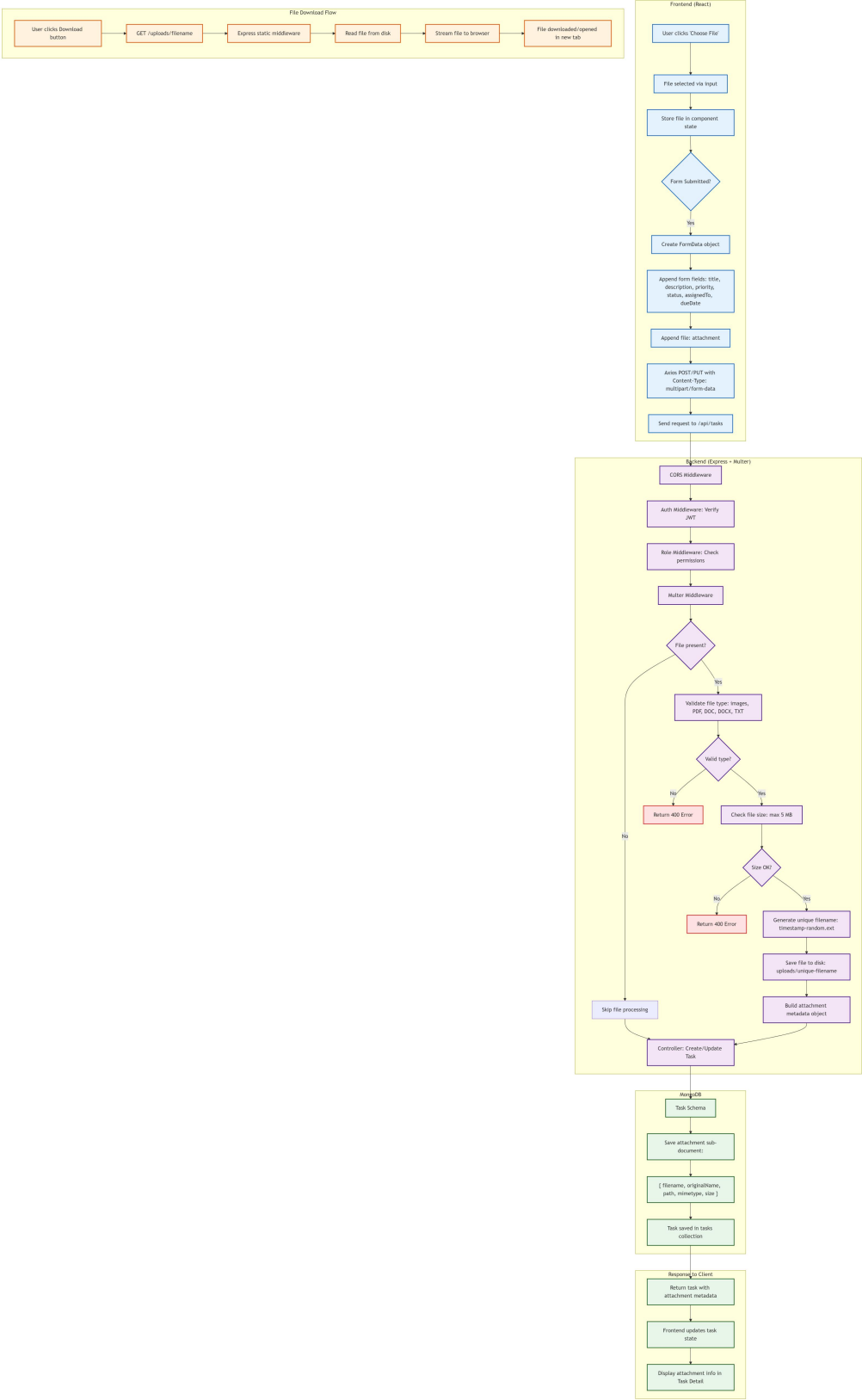
Authentication Flow



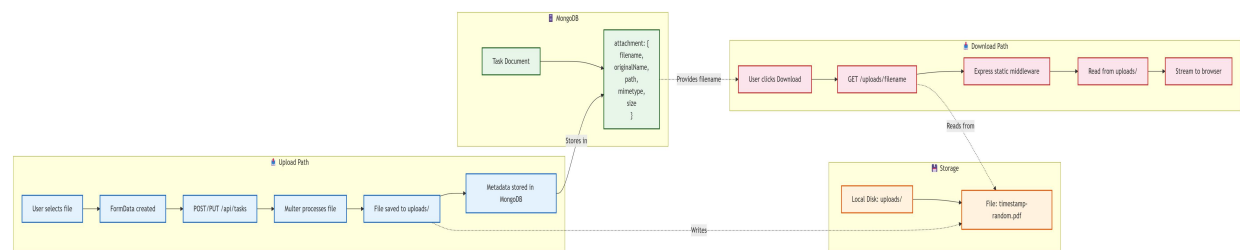
Task CRUD & File Upload Flow



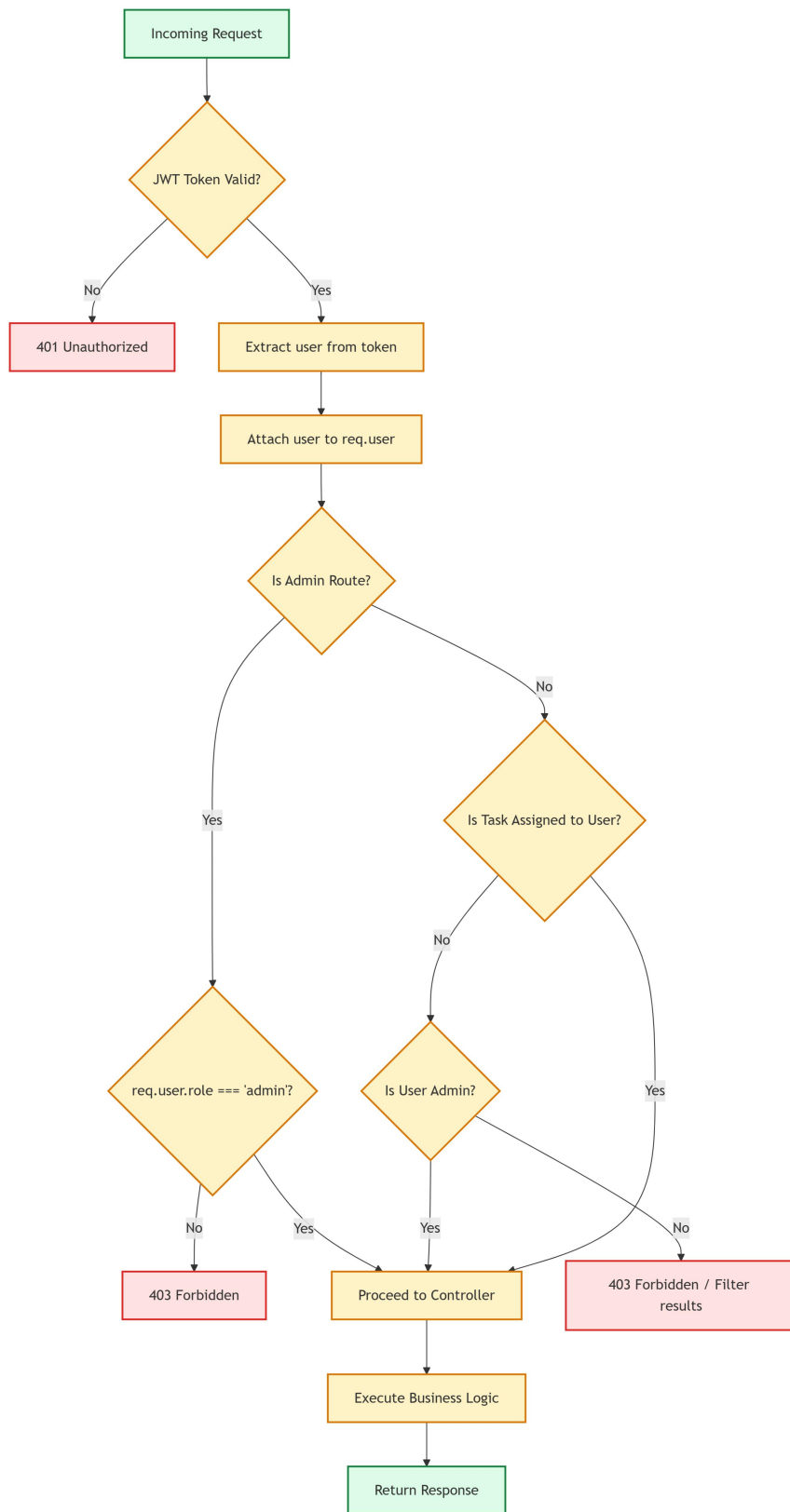
File Upload Flow Diagram



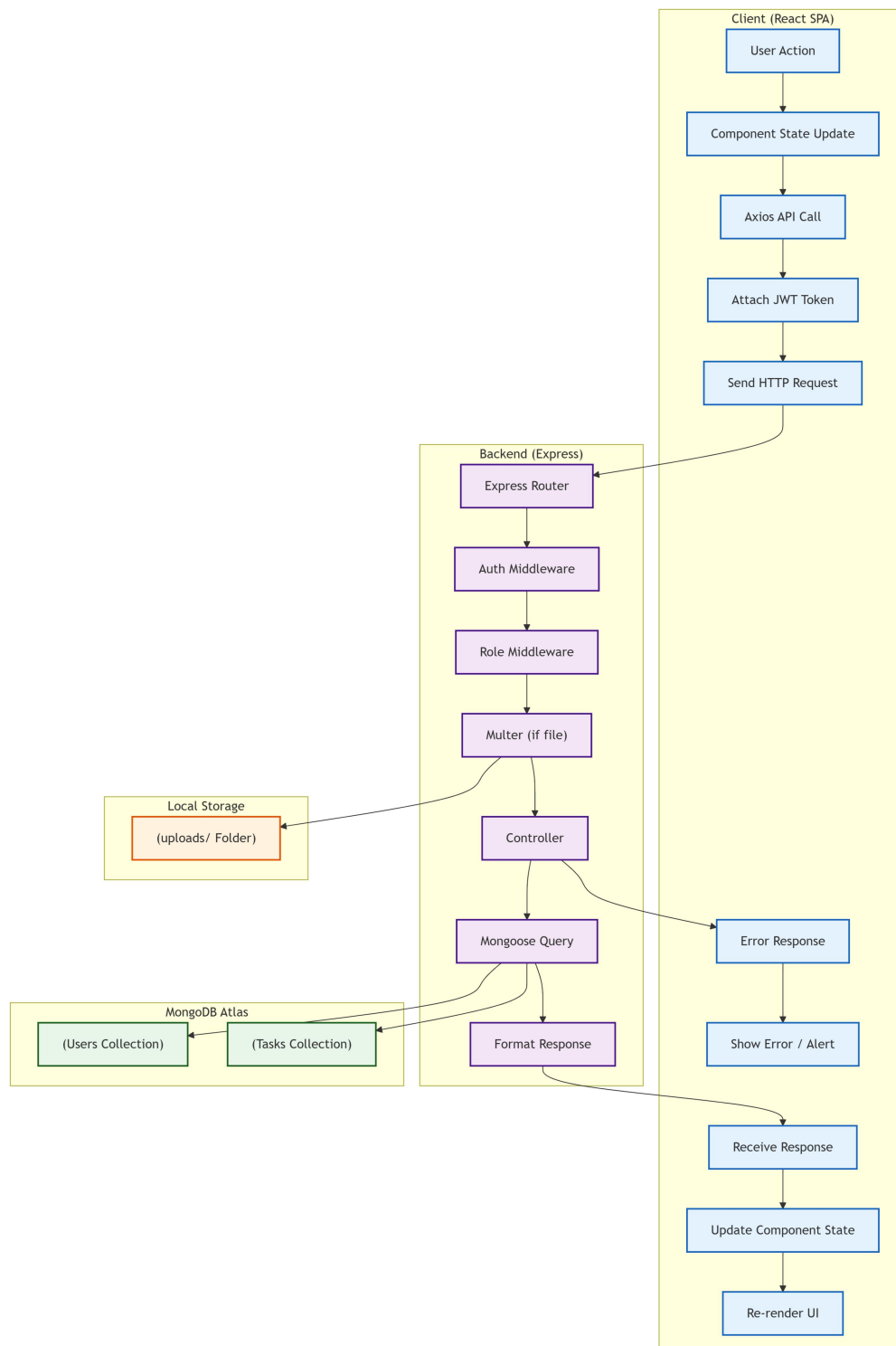
File Storage & Retrieval Architecture



Role-Based Access Control Flow



Data Flow Diagram



Deployment Architecture Diagram

